A Nishanth

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EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2028	8.79
The Velammal International School Panchetti	Class XII (CBSE)	2024	92.40%
RMK Patashaala	Class X (CBSE)	2022	97.00%

SCHOLASTIC ACHIEVEMENTS

Academic Excellence Award, IIT Hyderabad (2026) -- Awarded for securing the **highest GPA in the Biomedical Engineering department** during the **2024 academic year**.

PROJECTS

Intelligent Research Assistant using Retrieval-Augmented Generation (Applied AI & NLP)

Python, LangChain, Hugging Face, Sentence Transformers, ChromaDB, FAISS, Gemini API, Streamlit, FastAPI

Github: https://github.com/nishjuda/intelligence-research-assistant-rag

- Developed an **end-to-end Retrieval-Augmented Generation (RAG) platform** for semantic question answering and contextual summarization over unstructured PDF document collections.
- Designed a **metadata-aware retrieval pipeline** using **Sentence Transformers**, **ChromaDB**, and **FAISS**, enabling multi-document semantic search, page-level retrieval, citation-backed responses, and conversational document exploration.
- **Evaluated and optimized retrieval quality** using a custom benchmark of factual and summarization queries, experimenting with chunking strategies, overlap configurations, metadata filtering, and prompt engineering to improve answer grounding and contextual relevance.
- Built and deployed an interactive **Streamlit** application supporting document upload, conversational querying, page-specific and document-level summarization, retrieval visualization, and explainable AI through source attribution.

Chest X-Ray Pneumonia Detection System (Computer Vision)

(TensorFlow, Keras, FastAPI, OpenCV, NumPy, Pandas, Docker)

GitHub: https://github.com/nishjuda/chest-xray-pneumonia-detector

- Developed and deployed a **MobileNetV2 transfer learning**-based pneumonia detection system on **5,800+ chest X-ray images**, achieving **99% recall** and **0.89 F1-score** while minimizing false negatives for medical screening.
- Applied **data augmentation, class weighting, EarlyStopping, and learning-rate scheduling** to improve model generalization while minimizing false negatives in pneumonia detection.
- Integrated **Grad-CAM** to generate interpretable visual explanations by highlighting diagnostically relevant lung regions contributing to model predictions.
- Built a **FastAPI REST API** for real-time chest X-ray inference and **Dockerized** the application for reproducible, portable deployment across different environments.

RoadIQ - Road Accident Severity Prediction Platform

Python, XGBoost, Scikit-learn, Pandas, Streamlit, FastAPI, Docker, SQL, Plotly

GitHub: https://github.com/nishjuda/RoadIQ

- Developed and deployed an end-to-end accident severity prediction platform using **156K+ Los Angeles road accident records**, engineering **location, weather, temporal, and road infrastructure features** for real-time inference.
- Built a robust preprocessing pipeline with **missing-value imputation, frequency and one-hot encoding, temporal feature engineering, feature scaling, and data leakage removal** to improve model reliability.
- Optimized an **XGBoost classifier**, outperforming a **Random Forest baseline**, improving **Accuracy from 91.29% to**

92.47%, Recall from 72.3% to 77.9%, F1-score from 79.2% to 82.6%, and ROC-AUC from 0.956 to 0.963 through **hyperparameter tuning** and **threshold optimization.**
- Deployed the solution using **Streamlit, FastAPI, and Docker**, and integrated **SQL-based analytics** with interactive visualizations to explore accident severity trends across weather, infrastructure, and time.

SKILLS

Problem Solving: Solved **400+ DSA problems** and **100+ SQL problems** across LeetCode, GeeksforGeeks, and DataLemur.

Programming Languages:

Python, C++, SQL

Machine Learning & Deep Learning:

TensorFlow, scikit-learn, NumPy, Pandas, OpenCV

Generative AI & NLP:

LangChain, Hugging Face Transformers, Sentence Transformers, Retrieval-Augmented Generation (RAG), Prompt Engineering, Semantic Search, Information Retrieval

Vector Databases & Retrieval:

ChromaDB, FAISS, Embeddings, Metadata-aware Retrieval

Development & Deployment:

FastAPI, Streamlit, Docker, Git, GitHub

Generative AI APIs:

Gemini API

RELEVANT COURSES

Programming:

Introduction to Programming, Data Structures and Applications, Artificial Intelligence

Statistics & Mathematics:

Introduction to Probability, Introduction to Statistics, Elementary Linear Algebra

Mathematics:

Differential Equations, Calculus I & II

EXTRACURRICULAR

- **National Service Scheme (NSS)** - Participated in community service activities.

- **Football** - Active participant in recreational and inter-hostel football.

- **Strength Training & Fitness** - Maintain a consistent strength-training routine alongside academics.